

Tool-based Asymmetric Interaction for Selection in VR

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ABSTRACT

Mainstream Virtual Reality (VR) devices on the market nowadays mostly use symmetric interaction design for input, yet common practice by artists suggests asymmetric interaction using different input tools in each hand could be a better alternative for 3D modeling tasks in VR. In this paper, we explore the performance and usability of a tool-based asymmetric interaction method for a 3D object selection task in VR and compare it with a symmetric interface. The symmetric VR interface uses two identical handheld controllers to select points on a sphere, while the asymmetric interface uses a handheld controller and a stylus. We conducted a user study to compare these two interfaces, and found that the asymmetric system was faster, required less workload, and was rated with better usability. We also discuss the opportunities for tool-based asymmetric input to optimize VR art workflows, and future research directions.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; **Pointing**; **User studies**.

KEYWORDS

Asymmetric interaction, Virtual Reality, object selection, stylus

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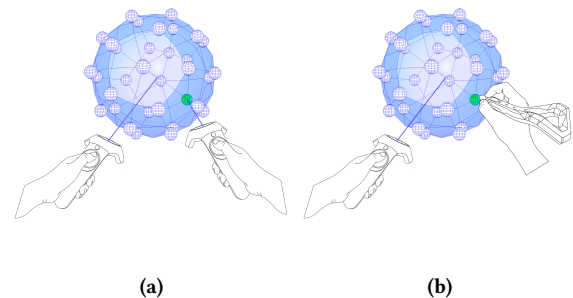


Figure 1: Symmetric interaction with Two-controllers (a) and Asymmetric interaction with Controller-stylus (b).

1 INTRODUCTION

Artist and designers use a variety of different input tools such as the mouse, keyboard, stylus or digital drawing pad to perform basic operations on the computer for 2D and 3D creative tasks, and asymmetric interaction plays a vital role in the design and art workflow. For example, a user might use a stylus in one hand and a keyboard in the other. This paper explores tool-based asymmetric interaction for 3D pointing/selection tasks in Virtual Reality (VR).

Using VR, artists can create and edit 3D objects through software, such as Tilt Brush [Google 2021] or Gravity Sketch [GravitySketch 2021]. In these applications a user can use handheld VR controllers to intuitively create or move 3D objects in a VR scene. However, the symmetrical interaction of two identical VR controllers designed for gameplay may be insufficient to perform the precise selection and efficient manipulation of a 3D model's points and surfaces as required in 3D modelling tasks [Arora et al. 2017; Velasco 2016].

This paper presents a tool-based asymmetric input method that enables artists to perform precise 3D object selection in VR (Figure 1b) and compares it with standard tool-based symmetric input (Figure 1a). The input method combines the standard controller of the HTC VIVE with the stylus of the Logitech VR Ink. The VR stylus provides 6 degrees of freedom (6 DoF) tracking, precise selection, pressure-sensitive input, and dynamic haptic feedback [Logitech 2020]. Our goal is to enable artists to use familiar real-world interaction techniques with asymmetric tools while in VR.

The main novelty of our work is that it provides one of the first user studies that compares 3D object selection in VR with a symmetric interface using two handheld controllers, to an asymmetric interface combining a stylus and a handheld controller. We aim to test the efficiency and usability of a tool-based asymmetric input system when performing basic 3D selection tasks in a VR environment. This is a first step in creating an intuitive asymmetric interface for artists and designers to use for real-time content creation in VR.

2 RELATED WORK

There has been considerable research on input devices and methods to provide precise interaction and natural user experiences in VR. For example, Rhijn et al. [Van Rhijn and Mulder 2006] created the Personal Space Station (PSS), a desktop near-field VR system, which used a pen-like input device with the ability to capture and correspond to the user's movement. Rhijn's research showed the great potential of using a spatially-aware controller for constructing a 3D object in the virtual environment. However, the user could not have a fully immersive and hands-free experience during the creation process.

Stylus input devices can also be used for precise input for VR interaction [Jackson 2020]. Previous research has found that stylus input is more reliable than finger and mouse for input (tapping and dragging) on tablets and phones [Annett et al. 2014; Batmaz et al. 2020; Chung and Shin 2015; Cockburn et al. 2012]. The stylus also provides a similar user experience as handwriting. Li et al. [Li et al. 2020] studied the appropriate pen position for holding a pen-like input device in a VR environment. They concluded that the TRE (tripod at the rear end) posture provides more considerable moving distances and more free tilt angles. They also found that a stylus had better performance than traditional wrist-based input in a poking task.

There are many stylus input devices for mobile phones or tablets on the market, and how to optimize and enhance these design tools for VR applications has been discussed. These studies ranged from position tracking systems to achieve angle-free input [Tian et al. 2007, 2008] to pressure-sensitive systems to increase input accuracy [Xin et al. 2012]. Grossman et al. [Arora et al. 2017] found that the lack of physical feedback and visual cues caused inaccurate operations in VR drawing. The Logitech VR Ink Stylus [Logitech 2020] is a novel stylus input device that provides analog force-sensitive controls and dynamic haptic feedback.

When using desktop 3D modeling software (such as Maya or ZBrush), artists often use different tools in each hand to create 3D objects. For example, the 3D sculpting-based modeling in ZBrush requires the user to use the stylus tablet and keyboard at the same time to handle both selection and carving. This suggests it is worth exploring the effect of asymmetric input also in a VR environment for 3D object creation tasks. Bai et al. [Bai et al. 2017] created an asymmetric bimanual interface for manipulating 3D virtual objects in a mobile VR application. They demonstrated a two-handed interface that combined the handheld controller with gesture input to provide natural and straightforward 3D manipulation for mobile VR. This study extends the existing research on VR asymmetric input systems by developing a tool-based asymmetric input system

using the Logitech VR Ink stylus with the HTC Vive handheld controller.

3 USER STUDY

3.1 System Design

In order to recreate the situation where two hands are using different tools simultaneously, we combine stylus input and a standard Vive controller in a tool-based asymmetric interaction. To evaluate our design, the experimental task (Figure 2) is based on the ISO 924-411 pointing task [ISO 1012] and used in prior work [Batmaz et al. 2020; Li et al. 2020]. We adapt this 2D task to 3D for simulating the situation where the user needs to select different virtual objects in VR. Thirty-one small white spheres are evenly distributed on the surface of a large translucent light blue sphere. The white spheres will randomly turn green in turn, and the user needs to select the green sphere by pointing at it with the right-hand controller and press the button on the controller to confirm the selection. The small sphere will turn back to white after the user confirms the selection. When all of the spheres have been selected once, the task is completed (Figure 3).

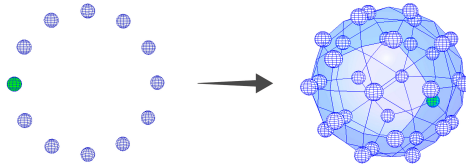


Figure 2: ISO 924-411 pointing task (left) and modified task design (right).

3.2 Experimental Design

We conducted a pilot study to evaluate the usability and user preference of the proposed tool-based asymmetric interaction method. The experiment had two conditions. First, a controller-stylus condition where the participants used an asymmetric input method consisting of a Logitech VR Ink with a standard Vive controller to accomplish the task. The second condition was the two-controllers condition where the participants used symmetric interaction with two standard Vive controllers (Figure 3).

3.2.1 Participants. A total of 12 participants (4 female, 8 male; age: 20-35) were recruited for the experiment from the university campus, and all of them had VR experience. Each participant had normal or corrected normal vision, and all had the same dominant hand (right hand). The experiment used a within-subject design, so participants were required to perform the same 3D object selecting task in the two different conditions, in a counterbalanced order.

3.2.2 Procedure. In the VR environment the user sees the large blue sphere connected to their left-hand controller. During the tasks, the user can move and rotate the left-hand controller to perform the corresponding operations on the light blue sphere. Once the

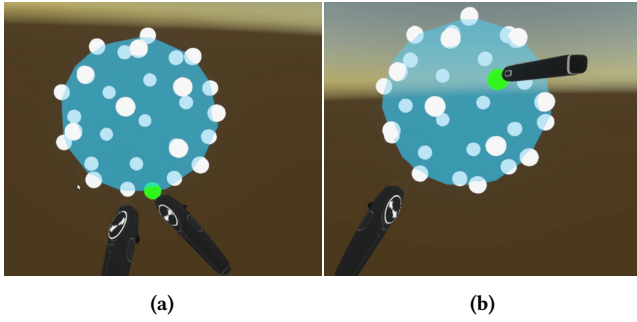


Figure 3: Screen shoots input methods with Two-controllers (a) and Controller-stylus (b).

task starts one of the small white spheres on the blue sphere randomly changes its colour to green and the user has to touch the green sphere with their right hand using either the controller (Two-controllers condition) or the stylus (Controller-stylus condition). The time and error for selection are logged automatically by the system.

Each participant has 5 minutes to read the instructions and get familiar with the basic manipulation of the controllers before the experimental trials. After finishing each condition the participant needs to complete the System Usability Scale (SUS) [Bangor et al. 2008], Simulator Sickness Questionnaire (SSQ) [Walter et al. 2019], and NASA-TLX [Hart and Staveland 1988] surveys.

3.2.3 Apparatus. We used the HTC Vive Pro attached to a PC with an NVIDIA GTX1080 graphics card and an Intel i7 processor. The handheld controllers were the standard Vive controllers. The stylus input device was a Logitech VR Ink, and its tracking system uses the SteamVR 2.0 sensors, being compatible with HTC Vive base stations. We used the Unity3D 2019 to modify the open-source program OpenBrush to create the 3D object manipulation task in VR.

4 RESULTS

4.1 Performance

4.1.1 Task Completion Time. The mean task completion time for each condition is shown in (Figure 4 left). The differences of task completion time for the two-controllers condition and controller-stylus condition were normally distributed, as assessed by Shapiro-Wilk's test ($p = .408$). Based on the paired-samples t-test, we found that participants were quicker to finish the task when using the controller-stylus asymmetric input system ($M = 49.600$ sec, $SE = 3.183$) in comparison to the two-controllers symmetric input system ($M = 57.798$ sec, $SE = 3.235$). The controller-stylus condition elicited a statistically significant decrease in task completion time compared to the two-controllers condition, $M = -8.19$ seconds, 95% CI [-15.512, -0.883], $t(11) = -2.467$, $p = .031$.

4.1.2 Error Rate. We collected the number of errors for the selection and used a paired-samples t-test (Figure 4 right) to compare error rate between the two conditions. There were no outliers in the data and the data were normally distributed ($p = .188$). There are no statistically significant mean difference ($p = 0.791 > 0.05$) of

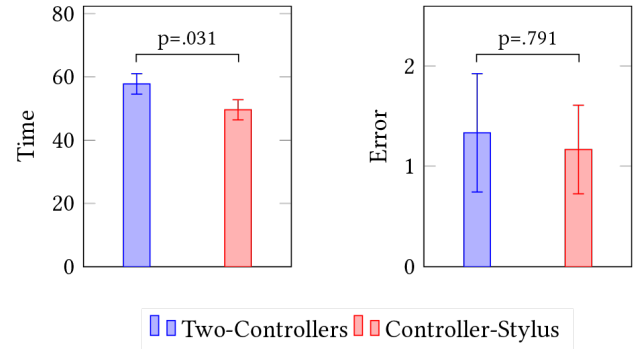


Figure 4: Performance results with task completion time in seconds (Error bar: +/- SE) (left) and Errors of the selection tasks (Error bar: +/-SE) (right).

error rate between the controller-stylus ($M = 1.167$, $SE = 0.441$) and two-controllers ($M = 1.333$, $SE = 0.589$).

4.2 Usability

4.2.1 System Usability Scale (SUS). We used the System Usability Scale (SUS) questionnaire to measure the usability of both systems. The data were not normally distributed as assessed by Shapiro-Wilk's test ($p = .04$). Thus, the Wilcoxon signed-rank test was conducted to determine the usability of each system. The difference scores were approximately symmetrically distributed, as assessed by a histogram with superimposed normal curve. The controllers-stylus input elicited a statistically significant increase in SUS scores compared to the two-controllers input, $z = 2.818$, $p = .005$. A Wilcoxon signed-rank test determined that there was a statistically significant increase in SUS score ($Mdn = 9.375$) when subjects used the controller-stylus input ($Mdn = 91.250$) compared to the two-controllers input ($Mdn = 81.875$), $z = 2.818$, $p = .005$.

4.2.2 Simulator Sickness Questionnaire (SSQ). Simulator sickness was measured by the SSQ survey. There were no outliers in the result of SSQ, and the data were normally distributed ($p = 0.50$). There are no statistically significant mean difference ($p = 0.209 > 0.05$) of SSQ between the controller-stylus ($M = 11.220$, $SE = 4.269$) and two-controllers group ($M = 15.750$, $SE = 4.403$) conditions.

4.2.3 NASA-TLX Subjective Work Load. The perceived workload of each interface was measured by using the NASA-TLX survey, and the results of six scales were analyzed using a paired sample t-test for comparisons separately (Figure 5). In terms of *Mental Demand* (MD), the controller-stylus condition elicited a statistically significant decrease in Mental Demand compared to the two-controllers condition, $M = -6.833$, 95% CI [-13.180, -0.486], $t(11) = -2.37$, $p = .037$. In terms of *Physical Demand* (PD), the controller-stylus condition elicited a statistically significant decrease in Physical Demand compared to the two-controllers condition, $M = -11.833$, 95% CI [-17.138, -5.029], $t(11) = -4.029$, $p = .002$. In terms of *Temporal Demand* (TD), the controller-stylus condition elicited a statistically significant decrease in Temporal Demand compared to the two-controllers condition, $M = -10.917$, 95% CI [-21.660, -0.173], $t(11) = -2.237$, $p = .041$.

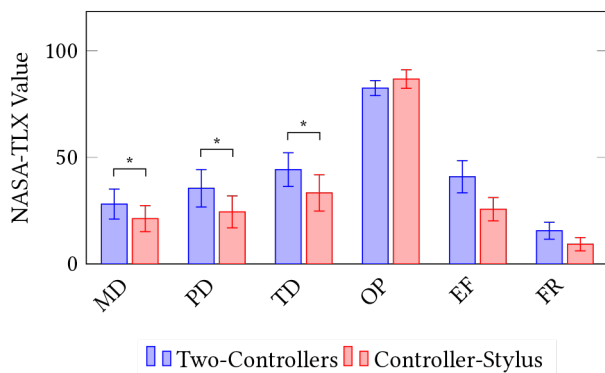


Figure 5: NASA-TLX Results. The items with significant differences are indicated by using an asterisk.

.047. Regarding *Performance* (OP), there was no statistically significant mean difference ($p = 0.36 > 0.05$) in performance between the controller-stylus group ($M = 86.750$, $SE = 4.356$) and two-controllers group ($M = 82.500$, $SE = 3.513$). Regarding *Effort* (EF), there was no statistically significant mean difference ($p = 0.08 > 0.05$) in Effort between the controller-stylus group ($M = 25.667$, $SE = 5.464$) and two-controllers group ($M = 40.917$, $SE = 7.547$). For *Frustration* (FR), there was no statistically significant mean difference ($p = 0.065 > 0.05$) of Frustration between the controller-stylus group ($M = 9.25$, $SE = 3.124$) and two-controllers group ($M = 15.583$, $SE = 3.976$).

4.3 Discussion

The study results show that the controller-stylus asymmetric interaction significantly improves the time required to complete the selection task with the same error rate compared to the traditional two-controllers symmetric interaction system. This is likely due to the fact that Logitech VR Ink has a shape closer to a pen, which enhances the user experience and precision of operation. Additionally, the interaction between the right hand holding the pen and the left hand manipulating the object is closer to the real workflow. The results also showed that users felt that the controller-stylus asymmetric interaction system had higher overall usability than the two-controllers symmetric interaction system since the asymmetric system ($91.25 > 90$) has an overall higher SUS score than symmetric system ($81.875 < 90$). In addition, based on the author's observation, some participants would accidentally cause a collision between two controllers when using two-controllers input. Two of the twelve participants had design backgrounds and both said, "If I had to choose between the two systems, I would be very happy to use the controller-stylus input to do some 3D design tasks in VR."

As a pilot study, our study could be improved in the future. The participants in this study were relatively experienced in using VR, and only two of them were engaged in art or design-related jobs. Future experiments will recruit more artists with experience in using desktop design tools to collect their experiences and opinions about the system. We also only measured performance and usability on a simple selection task. In the future it would be good to use

a range of more realistic tasks that use a combination of virtual object selection and manipulation.

4.4 Conclusion and Future Work

Through our preliminary study, we found that the controller-stylus asymmetric interaction can reduce task completion time and improve usability with less workload compared to two-controllers symmetric interaction while maintaining accuracy. In the future, we plan to integrate a data glove and drawing board into the tool-based asymmetric interaction system to provide intuitive gesture input, tangible feedback, and a more natural user experience. There are many other asymmetric combinations of input devices that could also be explored and evaluated.

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